Zach Deitch

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Education

Dickinson College, Carlisle, PA — Aug 2023 – May 2027
B.S., Mathematics; Minor in Computer Science
GPA: 3.99/4.00 | Phi Beta Kappa | Alpha Lambda Delta | Dean's List, All Semesters
Relevant Coursework: Artificial Intelligence (Fall 2026), Machine Learning (Spring 2027), Probability, Mathematical Statistics, Linear Algebra, Real Analysis, Algorithms
Activities: Squash Club (No. 1 ranked player); Badminton Club

Research Experience

Student Researcher, Mechanistic Interpretability — Jun 2026 – Present
Deep Neural Network Understanding Lab, Dickinson College, Carlisle, PA

- Reverse-engineered how attention-only GPT-style transformers (4, 6, and 7 layers, RoPE) trained on autocorrelated sequences infer an unknown lag from context.
- Built a path-decomposition framework showing early layers form a delay-coordinate representation through fixed-offset heads, while the final layer computes data-dependent lagged cross-products to select the predictive offset.
- Derived an exact algebraic decomposition of attention scores into routed scalar-product and bias terms, plus MSE-optimal predictors used as oracle baselines.
- Validated causally with activation patching, mean ablation, sign-flip tests, feature transplantation, and two-lag distribution shifts; programming final-layer attention to an arbitrary offset controls predictions, reaching the oracle floor exactly at the true lag and near-optimal error at lags outside the training range.
- Drafting a paper: *Dynamic Lag Retrieval in Small GPT-Style Transformers Trained on Autocorrelated Sequences.*

Student Researcher, Machine Learning and Optimization — Jan 2026 – Present
Dickinson College, Carlisle, PA

- Developed machine-learning-guided construction and local-search heuristics for the bounded multidimensional integer knapsack problem; benchmarked thousands of simulated instances against exact Gurobi solutions, reducing mean optimality gap to 1.00% from 1.15%.
- Building a neural classifier to predict effective formulations for binary quadratic programs.

Industry Experience

AI Engineer Intern — Jun 2026 – Present
Dickinson College, Carlisle, PA

- Proposed and secured IT support for Benny, a retrieval-augmented campus assistant spanning 20+ domains over structured data and 4,600 indexed web chunks.
- Built hybrid retrieval combining BM25 and dense search with reciprocal rank fusion and a cross-encoder reranker; added contextual retrieval at ingest to fix lookalike failures.
- Built an evaluation harness benchmarking 140 questions across 14 categories, achieving 92% correct-source retrieval.

Data Science Intern — Feb 2026 – May 2026
Kudu Investment Management, LLC, Remote

- Selected as one of five students to build a platform ranking 30,000 asset management firms, combining NLP classifications of SEC filings with growth and peer-relative metrics.
- Built an interactive application tracing the factors behind each ranking; presented the platform to Kudu's CEO.

Technical Skills

Machine Learning: PyTorch, transformer architectures, TransformerLens, activation patching and ablation, scikit-learn, retrieval-augmented generation, cross-encoder reranking

Mathematics: Probability, statistical inference, time-series analysis, linear algebra, optimization, Gurobi

Programming: Python (NumPy, pandas), SQL, R, Java, Git, Linux, Docker